
LYCANS AERODESIGN TEAM

PADA Report

Group Number: 02

1 Conceptual Design

1.1 Mission Requirements and Design Constraints

Mission Objective is to design a small unmanned aerial vehicle (UAV) resembling the Powered Autonomous Delivery Aircraft, aka PADA. The aim of the project is to simulate both aircraft design and manufacturing processes, which will follow when the season begins. Designing the aircraft means working on the wing, tail, fuselage, and electrical system.

The mission has very clear constraints:

Table 1: PADA UAV Mission and Design Constraints

Parameter	Constraint
Maximum Wingspan	25 in
Maximum Gross Weight	19.5 oz (including electrical system)
Payload	Empty Pepsi can (external mechanism)
Wing Airfoil Options	E423, S1223, GOE 165, or NACA 2412
Battery Energy Limit	$\leq 100 \text{ W h}$ (LiHV prohibited)
Takeoff Ground Distance	8 m to 12 m
Takeoff Angle of Attack	$< 35^\circ$ to 40°
Minimum Cruise Speed	25 m s^{-1}
Minimum Turn Bank Angle	$> 60^\circ$
Flight Altitude Range	50 m to 100 m

1.2 Selected Aircraft Configuration

1.2.1 Airfoil Selection

The chosen airfoil for the PADA's wing is the S1223 Airfoil, this is based on a trade study as seen below:

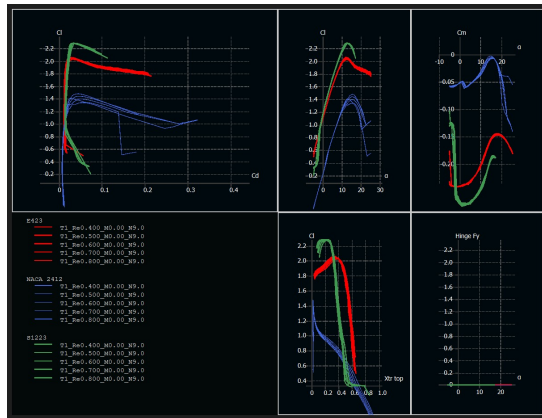


Figure 1: Airfoil Analysis and Comparison

Parameter	$C_{l\alpha=0}$	$C_{l,max}$	$\alpha_{C_{l,max}}$	$C_{d,min}$	$C_{m\alpha=0}$	Stall Variation
	A	A	C	A	A	A
		B	B	B	B	B
			C	C	C	C
				D	D	D
					E	E
						F
No. of occurrences	5	5	5	3	1	2
Weight	23.81	23.81	23.81	14.29	4.76	9.52
Airfoil	S1223	NACA 2412	E423			
$C_{l\alpha=0}$	1	-1	0			
$C_{l,max}$	1	-1	0			
$\alpha_{C_{l,max}}$	1	-1	0			
$C_{d,min}$	-1	1	0			
$C_{m\alpha=0}$	-1	1	0			
Stall Variation	-1	0	1			
Sum (weighted)	0.4286	-0.5238	0.0952			

Figure 2: Airfoil trade study

1.2.2 Wing and Tail Configurations

1.2.3 Fuselage

2 Preliminary Design

2.1 Propulsion Constraint Analysis

The Constraint Analysis was made based on the Constraints of the mission during Take-off, Cruising, Landing, and Turning, the following is a graph that demonstrates the Analysis:

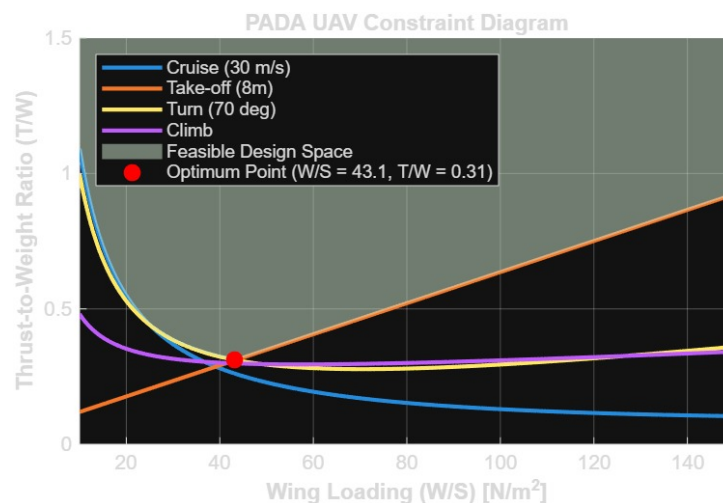


Figure 3: Propulsion constraint analysis diagram

The optimum point is highlighted where the Wing Loading is 43.1 and the Thrust to weight ratio is 0.31.

This analysis assumes an AR of 3.33 and a Climb rate of 3m/s which will be taken into consideration throughout the project, also R was given a value of 30m to have a smaller turn-radius.

2.2 Wing and Tail Sizing

2.3 Aerodynamic Modeling and Stability Analysis (AVL / XFLR5)

To evaluate aerodynamic performance and both static and dynamic stability, the sized aircraft was modeled utilizing XFLR5. The initial sizing and geometry were iteratively adjusted to

reach a converged, stable design. The mass and center-of-gravity (CG) distribution were also incorporated into the model.

2.3.1 Design Iterations and Configuration Selection

A short discussion is required to justify why the final configuration was selected over earlier iterations. Based on manufacturing and aerodynamic constraints, several major modifications were made:

- **Dihedral Angle:** A dihedral angle was initially considered for the main wing. However, it was cancelled in the final design because it proved too difficult to manufacture accurately. The final plane features a flat wing with no dihedral.
- **Tail Arm Length:** Attempts were made to increase the tail arm length to enhance stability margins. This modification increased the horizontal tail volume coefficient (V_h) drastically, resulting in overkill stability that would negatively impact pitch authority and maneuverability. Consequently, a normal tail arm length was maintained for the final aircraft.
- **Center of Gravity and Static Margin:** The CG was shifted significantly through multiple iterations to optimize longitudinal stability. Ultimately, the CG was fixed at a location that yields a final static margin of exactly 24%.

2.3.2 Analysis and Eigenmode Evaluation

Following the geometric adjustments, dynamic stability analysis was performed to extract the eigenmode values. The output plots and stability modes confirm that the aircraft demonstrates dynamic stability.

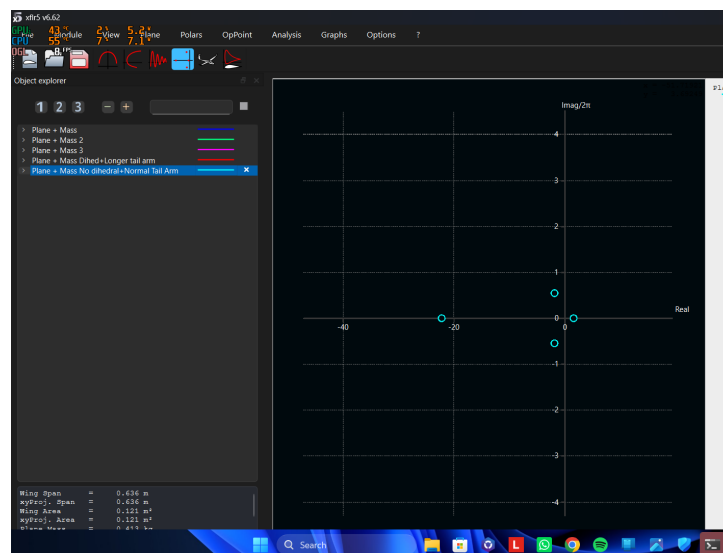


Figure 4: Dynamic stability eigenmodes and root locus plot from XFLR5. This plot confirms the damping and frequency of short-period modes, verifying the lateral dynamic stability of the final configuration.

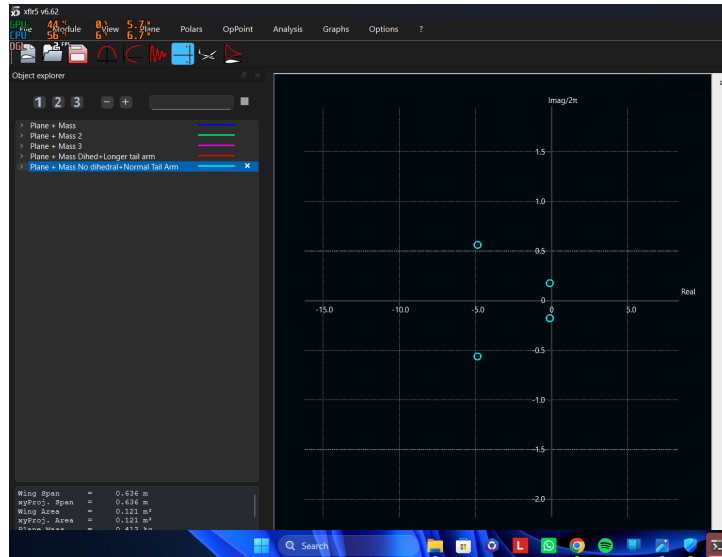


Figure 5: Dynamic stability eigenmodes and root locus plot from XFLR5. This plot confirms the damping and frequency of phugoid modes, verifying the longitudinal dynamic stability of the final configuration.

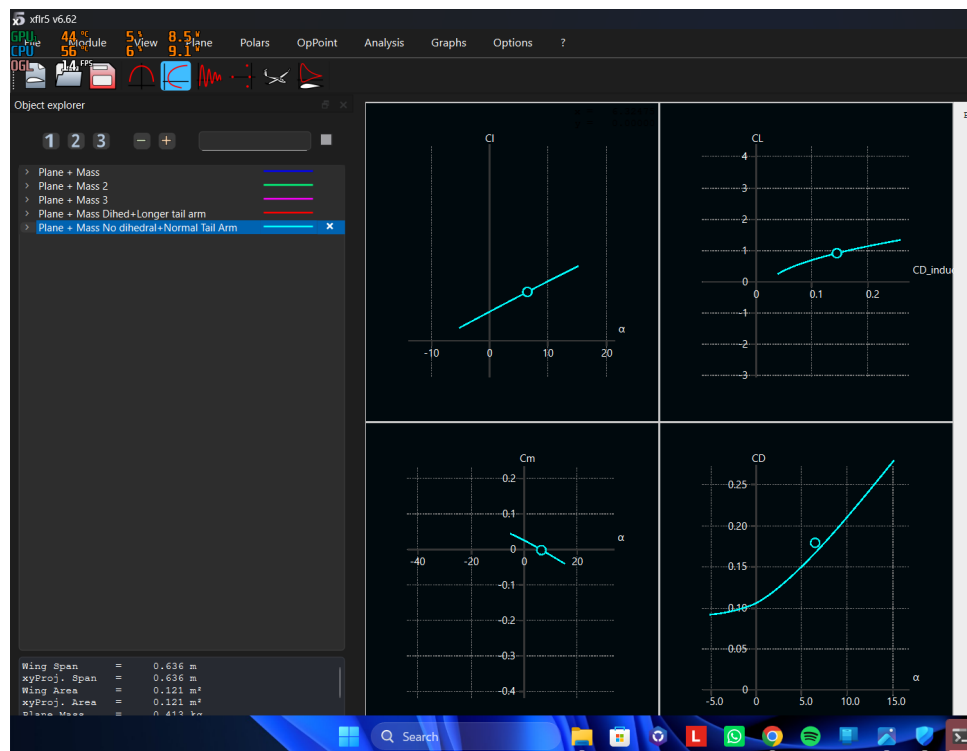


Figure 6: Aerodynamic performance polars. The charts illustrate the lift-to-drag ratios and stability derivatives across varying angles of attack for the final no-dihedral wing configuration.

2.4 Aircraft Design

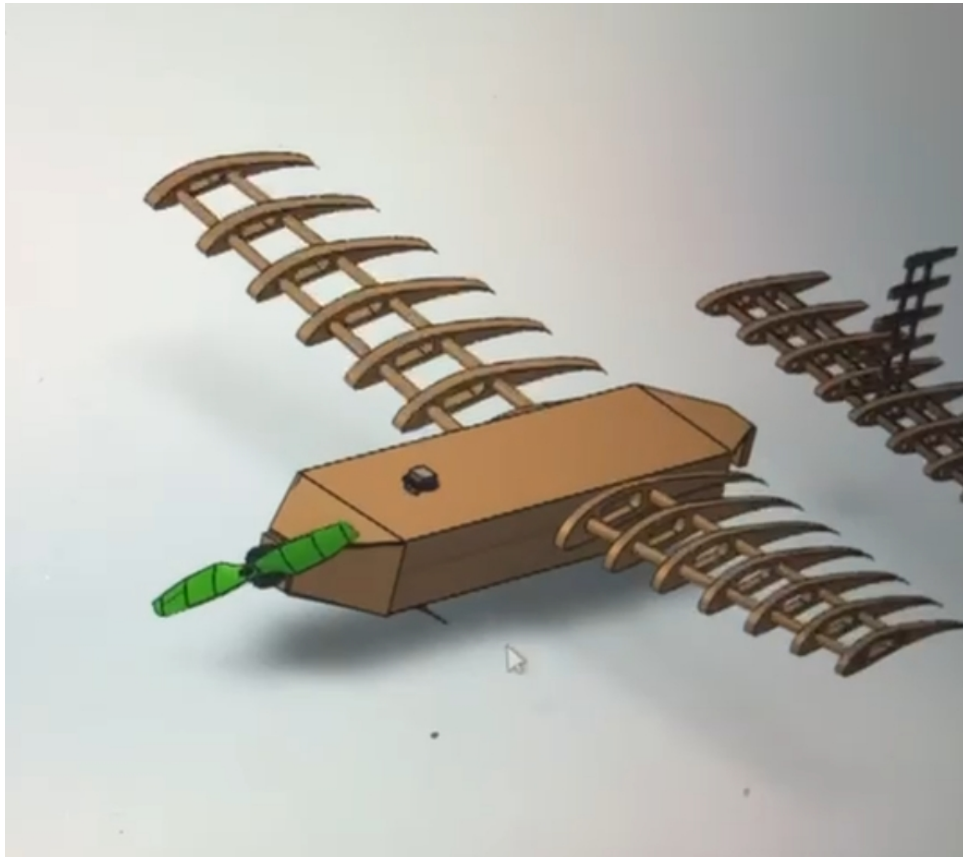


Figure 7: Solid works implantation

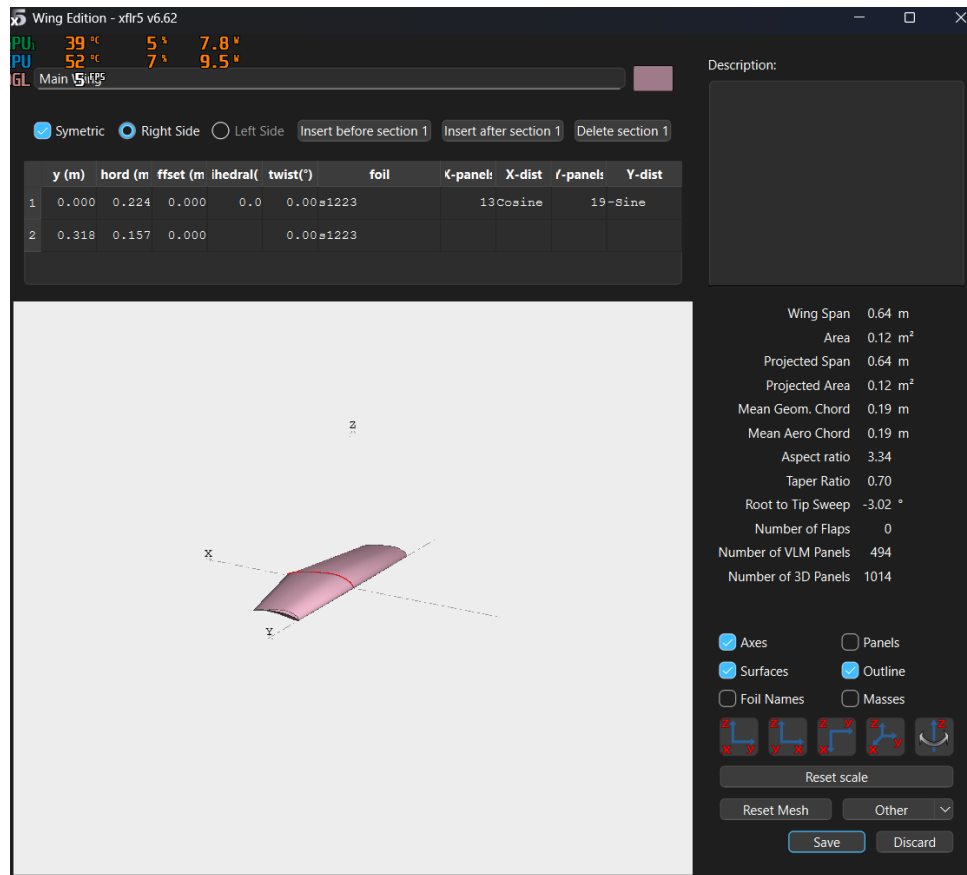


Figure 8: Final main wing configuration featuring zero dihedral for optimized manufacturability and structural simplicity.

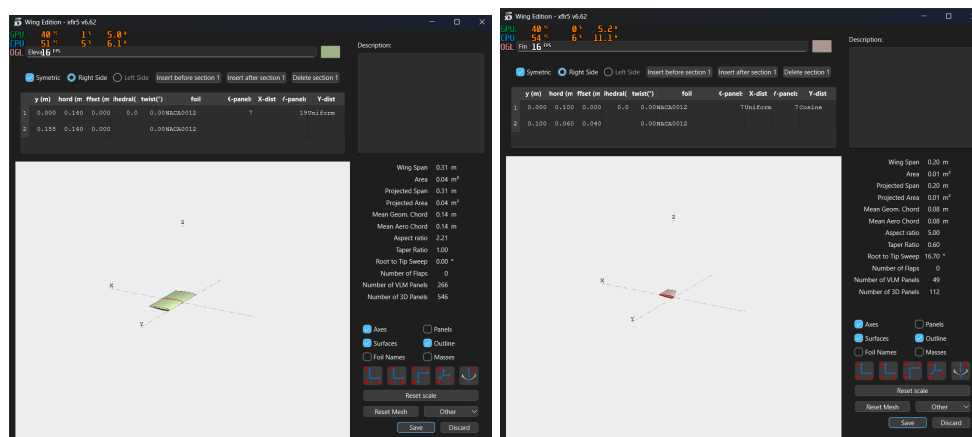


Figure 9: Final empennage design utilizing a normal tail arm length to maintain an optimal horizontal tail volume coefficient (V_h) without inducing excessive stability.

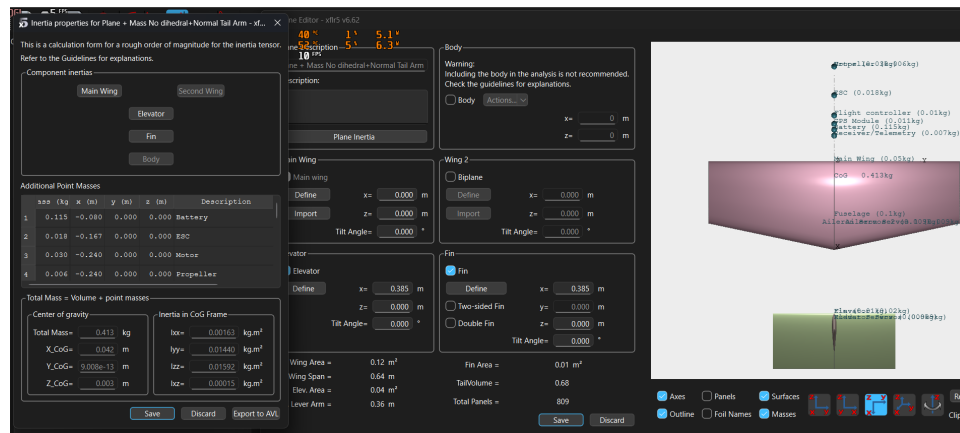


Figure 10: Final Center of Gravity (CG) placement on the aircraft, yielding a 24% static margin for balanced static stability.

2.5 Electrical and Avionics Design

2.6 Autonomous System and Software

2.7 Payload Release Mechanism

2.7.1 Final CAD design and release principle.



Figure 11: All items needed for the payload release mechanism

- Movable Arm
- Fixed Arm
- Back Support

How does it work? A fifth servo was added to move the controllable arm outwards causing the payload to drop but two arms aren't enough to hold the payload so a third arm was added to the back of the payload to maintain its stability. This mechanism was strategically mounted directly under the final CG to maintain structural balance and ensure the 24% static margin remains completely unaffected during the payload drop.

2.8 Material Selection

When evaluating candidate materials for the airframe, the primary selection criteria were weight, durability, cost, and availability. Balsa wood was selected as the primary material for all struc-

tural components. Balsa provides an excellent strength-to-weight ratio, ensuring the airframe is durable enough to withstand aerodynamic loads while maintaining a low mass to stay well under the maximum gross weight limit. Furthermore, balsa wood is highly cost-effective and was realistically the only viable, accessible material that suited both our manufacturing capabilities and the mission constraints.

3 Manufacturing Plan

The manufacturing strategy was designed around rapid prototyping and precise assembly using laser-cut balsa wood. The complete process is outlined as follows:

- **Material Sourcing:** Standard balsa wood planks, specifically dimensioned at 1000 mm × 100 mm × 5 mm, were procured for the airframe construction.
- **Digital Preparation:** The full 3D CAD assembly was translated into 2D manufacturing instructions. These vector files contained the exact layout for all necessary structural parts, including wing ribs, fuselage formers, and empennage surfaces.

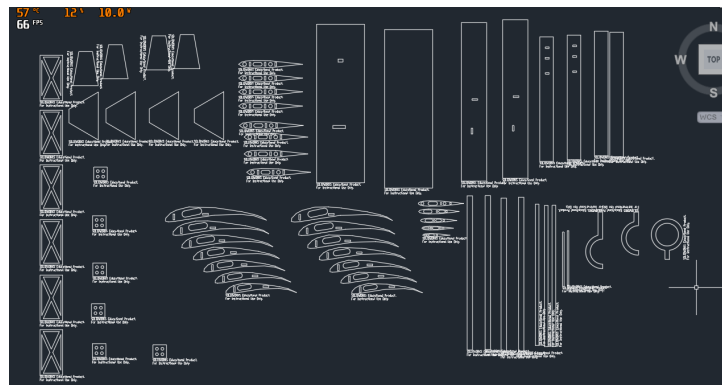


Figure 12: Full 2D manufacturing instructions

- **Laser Cutting:** The finalized CAD cutting instructions were sent to the laser cutting facility. Using a laser cutter allowed for extremely high-precision cuts, ensuring that all interlocking tabs and slots would fit together perfectly without requiring excessive manual sanding or modification.
- **Assembly:** After all parts were cut from the balsa planks, they were aligned and glued to form the final airframe structure, preceding the integration of the electrical systems and payload mechanism.